Appendix E
Performance Metrics

We adopt the evaluation protocol of the TravelPlanner benchmark [9] and report five metrics: *Delivery Rate*, *Commonsense Constraint Pass Rate* (micro and macro), *Hard Constraint Pass Rate* (micro and macro), and *Final Pass Rate*. Below we provide precise definitions, with concrete examples for transparency.

a) Delivery Rate: The percentage of runs that successfully produce a schema-valid itinerary within the step budget (30 steps in our setting). A run fails if it exceeds the step limit, falls into a dead loop, or generates an invalid schema.

b) Commonsense Constraint Pass Rate: This metric evaluates whether itineraries respect implicit feasibility rules, even when they are not explicitly stated in the user query. Following [9], we evaluate eight commonsense dimensions: (i) *Within Sandbox* (the plan does not hallucinate entities outside the environment), (ii) *Complete Information* (all key components such as accommodation and transportation are specified), (iii) *Within Current City* (daily activities occur in the correct city), (iv) *Reasonable City Route* (city transitions follow a logical sequence), (v) *Diverse Restaurants* (restaurant choices are not repeated), (vi) *Diverse Attractions* (attractions are not repeated), (vii) *Non-conflicting Transportation* (no contradictory modes such as “flight” and “self-driving” in the same trip), and (viii) *Minimum Nights Stay* (accommodations respect stay requirements).

c) Hard Constraint Pass Rate: This metric measures adherence to user-specified requirements, such as budget, accommodation rules (e.g., no pets or smoking), accommodation type (entire, private, or shared), cuisine preferences, and transportation constraints (e.g., “no flights”).

d) Micro vs. Macro pass rates: Let P denote the set of all evaluated plans and C_p the set of constraints applicable to plan $p \in P$. Define $\mathbf{1}_{\text{passed}}(c, p) \in \{0, 1\}$ as an indicator that plan p satisfies constraint c, and $\mathbf{1}_{\text{passed}}(C_p, p) = 1$ if *all* constraints in C_p are satisfied (else 0). Then:

$$\text{Micro Pass Rate} = \frac{\sum_{p \in P} \sum_{c \in C_p} \mathbf{1}_{\text{passed}(c,p)}}{\sum_{p \in P} |C_p|}.$$

$$\text{Macro Pass Rate} = \frac{\sum_{p \in P} \mathbf{1}_{\text{passed}(C_p,p)}}{|P|}.$$

Intuitively, the micro score measures success *constraint by constraint* across all plans, while the macro score is stricter, counting a plan only if it satisfies *all* of its applicable constraints.

e) Final Pass Rate: The proportion of plans that simultaneously (i) are delivered, (ii) pass all applicable commonsense constraints, and (iii) pass all applicable hard constraints.